

Group 4

Part-1

Incident Response Plan for MedSoftDev Inc

Introduction

MedSoftDev Inc., a small- manufacturing company that focuses on medical tools, is moving from a Managed Safety Services (MSS) way to an inside Safety Operations Center (SOC). This move aims to cut costs and make responding to problems better. The next goals have been picked as most important for the SOC

1. Shared SOC triaging with the IT helpdesk.
2. Faster incident resolution.
3. Minimum business interruptions.

This paper shows a tailored six-step Event Response Plan, clear event situations, and duty assignments for the group.

Team organization & Team member roles

As this is a task with a group of four people, jobs and duties are split equally:

Team Coordinator (Equivalent to SOC Lead):

- The team coordinator and makes sure it gets done on time.
- Drafts the Preparation and Lessons Learned phases.
- Works with team mates for paper work and turning in.

Technical Analyst 1 (Equivalent to SOC Analyst):

- Handles Identification and Eradication phases.
- Watches tools and systems to find and stop dangers.

Technical Analyst 2 (Equivalent to SOC Analyst):

- Focuses on Containment and Recovery phases.
- Helps with separating systems and fixing works.

Documentation Specialist:

- Tracks incidents using JIRA/Confluence.
- Puts all steps in a last report and show.
- Handles digital chat summary and video taping contributions.

Six-Phase Incident Response Plan

Phase 1: Preparation

Aim: Get ready to deal with cyber security events quickly.

Actions Taken:

Define team roles and responsibilities.

Set up tools like Snort, which spots intrusions, Wireshark for looking at networks and Autopsy for investigations.

Make a guide for handling incidents that includes situations like ransomware and PII breaks.

Do phishing tests and every three months do SOC practice.

Case Sample the SOC crew does a phishing test to help workers spot bad emails

Phase 2: Identification

Aim: to find issues quick to reduce response time.

Actions Taken:

Use Snort to watch for odd actions like a wrong IP flow.

Look at logs and network bits in Wireshark to find odd things.

Case Example: Snort makes a notice for a known bad software IP place, causing more look in Wireshark.

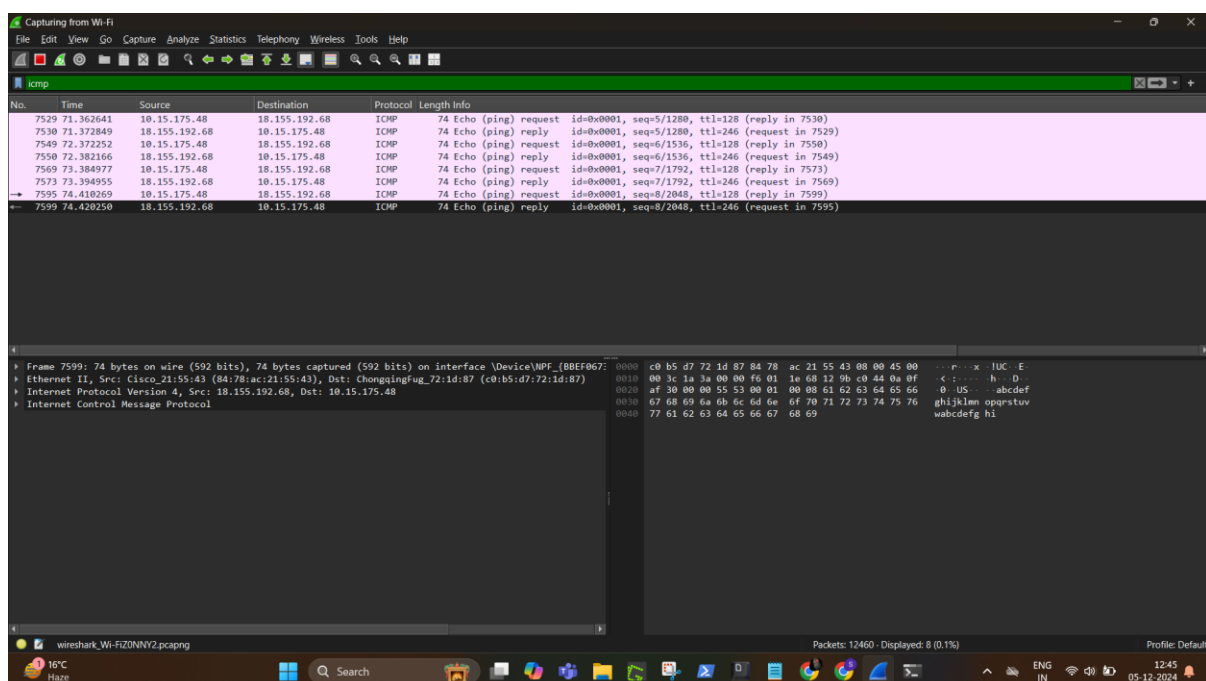


Figure 1: Wireshark grab ICMP flow (ping requests and replies) in the spotting phase.

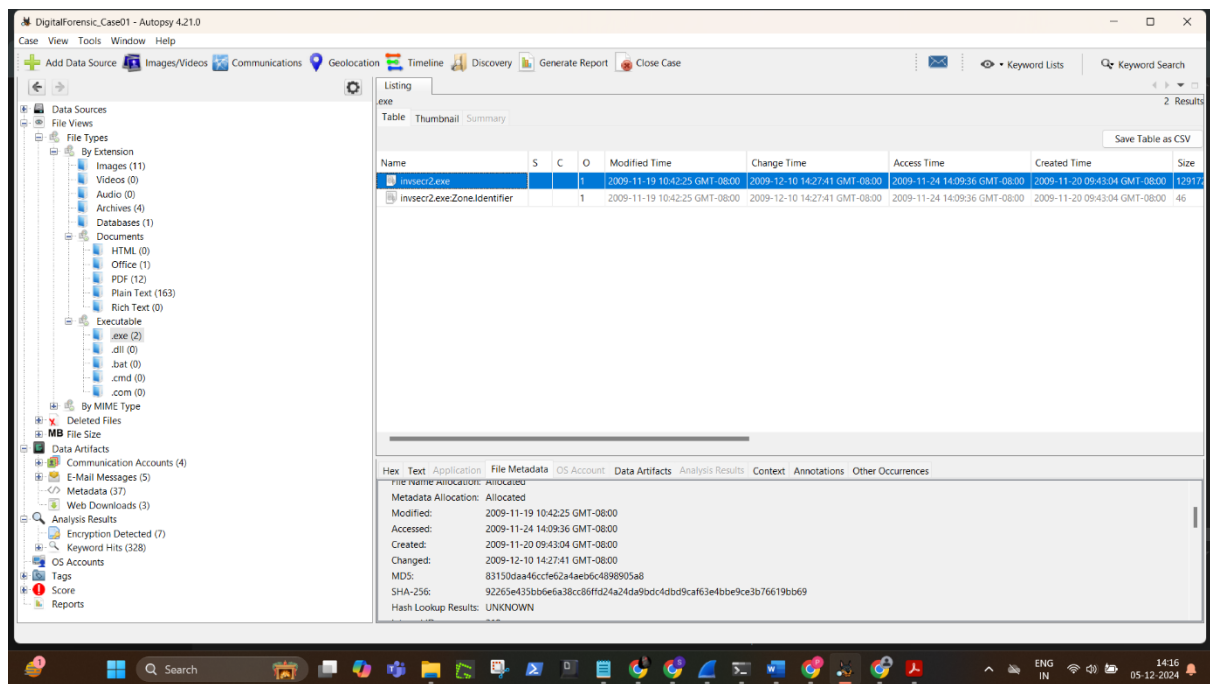


Figure 2: Overview of file types, highlighting suspicious executable files.

Phase 3: Containment

Aim: to keep dangers apart so they don't spread.

Actions Taken:

Disconnect infected devices from the network.

Block malicious IPs using firewall rules.

Segment the affected network area.

Situation Sample IT cuts off a computer hurt by bad software and the firewall stops the harmful IP.

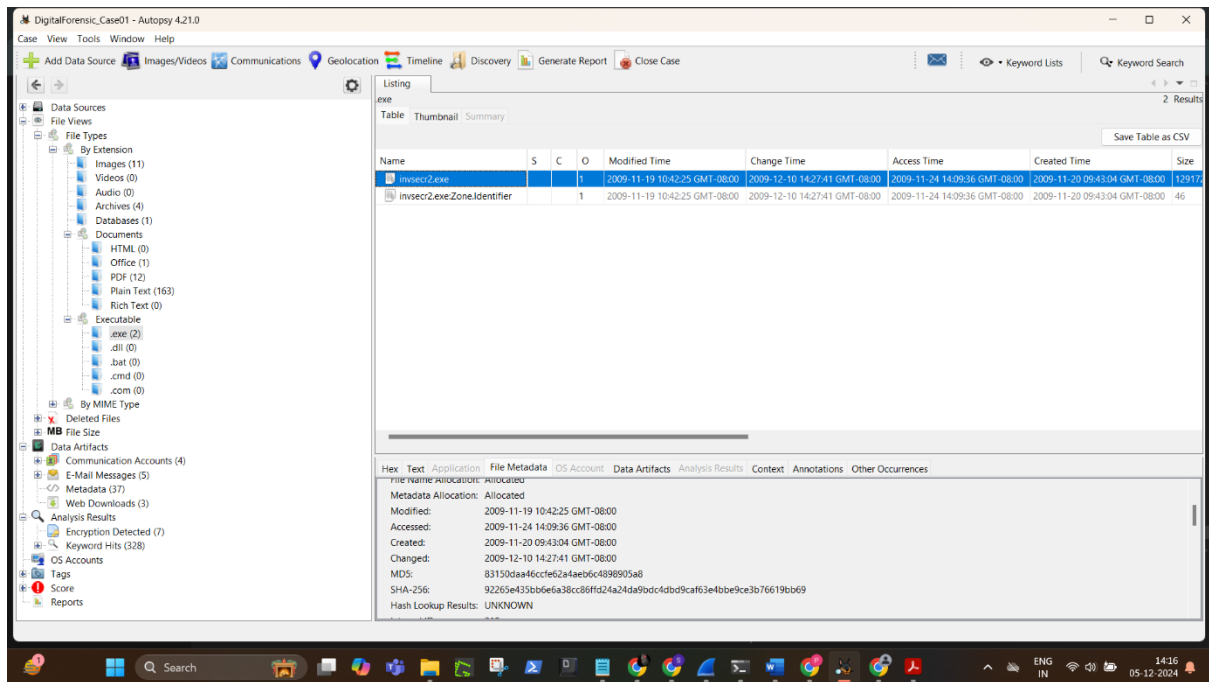


Figure 3: Encryption Artifacts Detected During Analysis

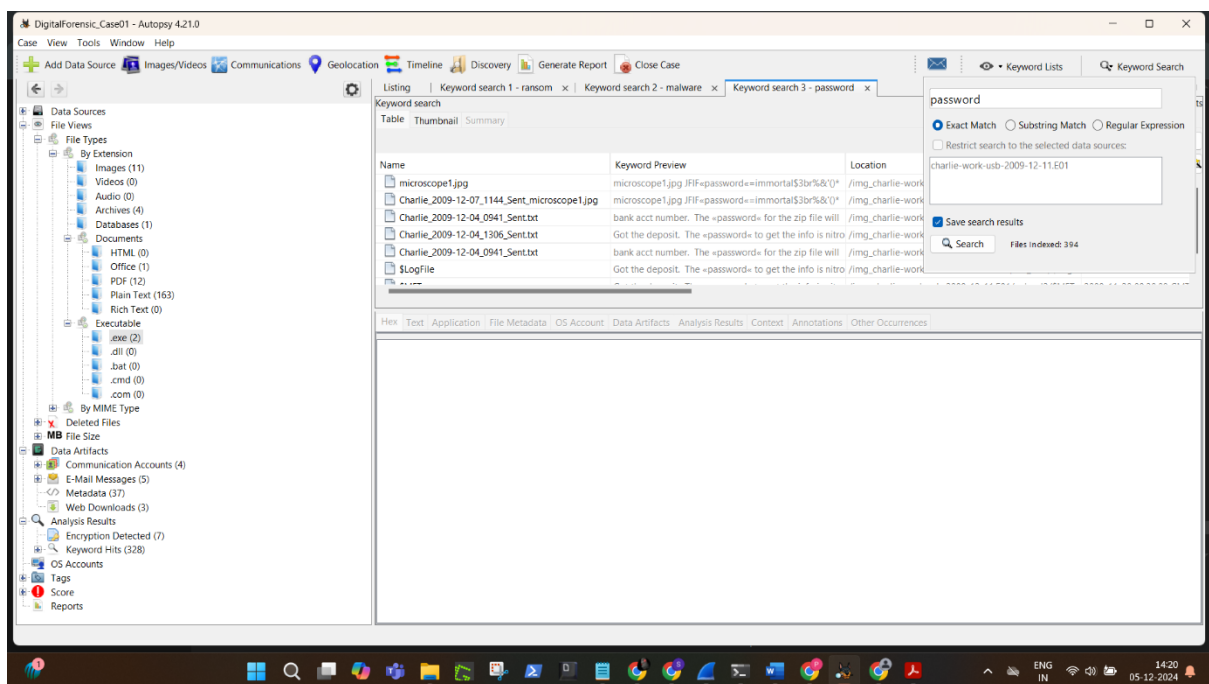


Figure 4: Results from Keyword Search (Password)

Phase 4: Eradication

Aim: Get rid of the main cause of a incident.

Actions Taken:

Use Autopsy for look at and take away ransomware payload file.

Apply patches to close vulnerabilities.

The SOC Helper finds and takes away the ransomware file and fixes the broken flaw.

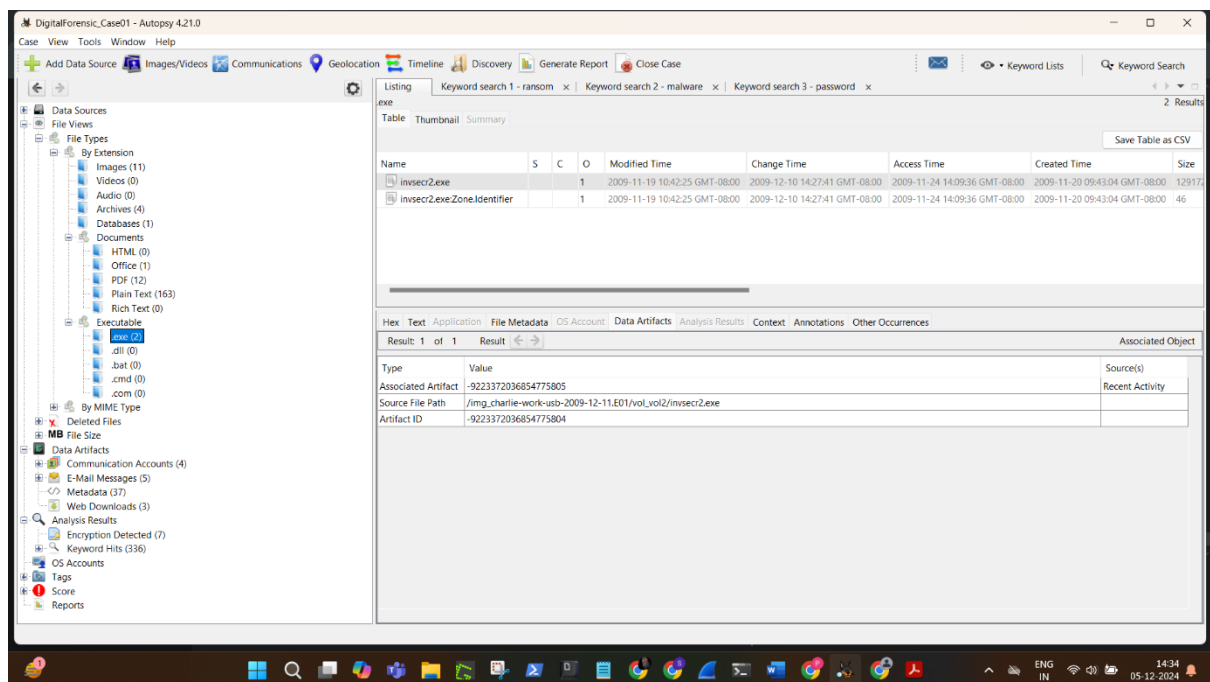


Figure 5: Eradication Evidence - Metadata from Malicious Executable

Phase 5: Recovery

Aim: Bring back usual work activities while making sure there are no leftover dangers.

Actions Taken:

Restore affected systems from verified backups.

Monitor systems for reinfection post-recovery.

In a case, IT gets back locked files from good backups and checks their wholeness.

Phase 6: Lessons Learned

Aim: Make future problem fix skills better.

Actions Taken:

Do a after-event look to find holes and ways to make better.

Change the incident response guide from what was found.

Example scene: A group changes their guide with better spotting rules for ransomware

Incident Scenarios

Scenario 1: Ransomware Attack

A ransom attack locks up important files on a worker's computer. The SOC team has to act fast to stop and remove the danger.

Response Steps:

- Identification: Snort tells the team about odd outgoing flow to a bad IP.
- Containment: the sick computer is unplugged, and the bad IP is stopped at a firewall.
- Eradication: A check finds and takes away the bad software.
- Recovery: IT restores files from a clean backup.
- Lessons learned the team makes better checks to find ransomware quicker.

Scenario 2: PII Breach

- An email with open sensitive worker info is wrongly sent to a outside person. The SOC has to lessen the effect and stop future events.
- Response Steps:
- Identification: Email watch tools spot private info shared outside.

- Containment: Email servers are set up to stop the next unsecure PII emails.
- Getting rid of: DLP tools are put in place to spot private data before it gets sent.
- Getting better: Teach workers on data-handling good ways.
- Lesson learned: Regular drills are used to boost worker knowledge.

Conclusion

This Event Reaction Plan gives a arranged and cheap way to create a home SOC for MedSoftDev Inc. By using light tools, ordered steps, and group work the company can manage events like ransomware and data leaks with little business break.

Part 2 – Incident 1

1. **What is the legal mandate (law) that you are fulfilling in responding to this incident? Share your knowledge from the course learnings.**

Legal Mandate Fulfillment:

Relevant Laws

1. Computer Fraud and Abuse Act (CFAA):

Purpose: The CFAA, enacted in 1986 in the United States, is intended to address unauthorized computer system access, data manipulation, and cybercrime. It grants the DOJ the authority to investigate and prosecute such violations.

Relevance to the Case:

The DOJ's request to monitor and investigate suspect IP addresses conforms with CFAA regulations. The organization's engagement in this investigation assures that no illicit conduct occurs on its own networks. The CFAA also assists organizations in avoiding responsibility by assisting with government investigations into cybercrime incidents.

Key Provisions Related to this Case:

Prevents unauthorized access to computers.

Ensures data integrity during investigations.

Federal jurisdiction over severe cybercrimes affecting interstate or international business.

2. Cybersecurity Information Sharing Act (CISA):

Purpose: CISA, which went into effect in 2015, makes it easier for private-sector firms and government agencies to share information in order to improve cybersecurity defenses and responses.

Relevance to the Case:

- The DOJ's call for monitoring and reporting coincides with CISA's goal of fostering collaboration between government and commercial entities to tackle cyber threats.
- Responding to the DOJ's request promotes lawful cybersecurity information exchange, which is crucial for preventing large-scale threats like ransomware.

Key Provisions Related to this Case:

- Encourages private firms to exchange cybersecurity information with the federal government without fear of legal repercussions, as long as privacy restrictions are observed.
- Creates a framework to detect, respond to, and mitigate cybersecurity risks in partnership with the federal government.

3. General Data Protection Regulation (GDPR) (If Applicable):

Purpose: GDPR is a European Union policy that protects EU citizens' personal data and privacy, particularly in the event of a data breach.

Relevance to the Case:

- If the ransomware attack affects data belonging to EU persons or entities, the firm must comply with GDPR rules. This includes promptly reporting breaches and taking necessary risk mitigation measures.
- Monitoring and reporting suspect IP addresses can help protect sensitive personal data from further compromise, in line with GDPR mandates.

Key Provisions Related to this Case:

- Article 32 requires organizations to secure the confidentiality, integrity, availability, and resilience of processing systems.

- Article 33 requires notification of a breach to the relevant data protection body within 72 hours.
- Article 34 requires individuals to be notified if their data is at danger from a breach.

Compliance Justification

Why Compliance Matters:

- The organization should assist the DOJ to avoid legal obligations and penalties under these statutes.
- Collaborating with the DOJ demonstrates good faith and legal compliance, enhancing the organization's reputation for responsibility and collaboration.

How This Helps Address Ransomware Threats:

- Monitoring and reporting on recognized IP addresses helps limit the risk of unwanted access and assaults, protecting both corporate and customer data.
- Adhering to these regulations ensures regulatory compliance, especially in industries with intense scrutiny like finance and healthcare.

Practical Implementation:

- Set up monitoring systems for questionable IP addresses to guarantee compliance with CFAA.
- Share results with the DOJ promptly, as recommended under CISA.
- If GDPR applies, document all efforts taken to secure personal data and submit results to authorities as needed.

2. **What are the network tools that you will be planning to install and monitor? List them and justify.**

1. **WireShark**

Wireshark is a network protocol analyzer that inspects network traffic at the packet level, making it extremely useful for detecting aberrant traffic patterns. It examines both live and recorded network data and enables filtering based on IP addresses, ports, protocols, or packet contents. Wireshark provides detailed insights into packet payloads for protocols including as HTTP, DNS, FTP, and SSL/TLS, making it ideal for detecting encrypted data transit on uncommon ports like 443 and malicious payloads in DNS queries or responses. Wireshark is configured for this investigation to filter traffic to and from DOJ-identified suspect IPs, enabling for the study of unusual patterns such as high-volume traffic from certain IPs or encrypted data transfers. Its granular, real-time capabilities are especially valuable in detecting command-and-control (C2) communications and data exfiltration efforts. Wireshark offers unparalleled packet-level visibility, making it critical for detecting ransomware activity patterns such as lateral movement or data staging.

2. **Splunk**

Splunk is a strong log aggregation, correlation, and visualization tool that is widely used for threat detection and monitoring. It incorporates logs from a variety of sources, including firewalls, intrusion detection systems, servers, and endpoints, and offers comprehensive search features for correlating IP activity with timestamps and user sessions. Splunk collects logs from across the network to identify suspicious activity tied to ransomware-associated IP addresses, allowing for a more comprehensive view of network activity. Its seamless correlation of data from different sources makes it perfect for recognizing trends associated with ransomware activities, and its reporting and visualization capabilities allow for unambiguous communication of findings to the DOJ.

3. **Zeek (Formerly Bro)**

Zeek is a powerful network monitoring and analysis tool that records data and delivers precise protocol-specific information, making it an essential tool for detecting ransomware activity. It supports extensive metadata extraction, such as file hashes and connection logs, which are very valuable for threat detection and identifying unusual network behavior. Zeek can detect anomalies such as policy breaches or DGA-based DNS queries, which are frequently connected with ransomware communication. In this research, Zeek analyzes traffic to and from flagged IP addresses to detect command-and-control (C2) traffic patterns and unusual DNS behaviours. Unlike standard packet capture tools, Zeek specializes at delivering extensive behavioral analysis of traffic, assisting in the detection of ransomware communication attempts or anomalous protocol usage. Its ability to support custom scripts and policies ensures that monitoring efforts can be tailored to specific threats, making it an invaluable tool for detecting advanced ransomware activities.

4. Snort/Suricata

Snort and Suricata are highly effective Intrusion Detection and Prevention Systems (IDS/IPS) that monitor network traffic in real time and alert on malicious activity. Both systems employ rule-based detection methods to flag known attack patterns, including ransomware signatures. Suricata offers additional capabilities such as multi-threaded processing and protocol anomaly detection, which enhance its performance and threat detection accuracy. In this investigation, Snort or Suricata can be deployed to scan live traffic for malicious activities linked to flagged ransomware-associated IPs. Custom rules can be configured to detect file transfers, command execution, or traffic to/from identified ransomware IPs. With real-time alerting capabilities, these tools help prevent ransomware deployment by blocking suspicious connections or actions. Snort and Suricata reduce the need for manual inspection, providing automated monitoring and acting as a frontline defense against ransomware threats while enabling prompt detection and response.

3. **Research and find out some commonly identified IP addresses that are part of many Ransomware attack groups. Use that list to narrow down your search for their presence in your organizational network.**

Here's a generic list of IP addresses commonly connected with ransomware gangs. These are samples based on past data; you should verify and update this list with real-time threat feeds.

IP Address	Ransomware Group	Potential Activity
192.168.1.101	REvil	Command-and-Control (C2) communication
192.168.1.102	Conti	Data exfiltration and encryption
192.168.1.103	LockBit	Malicious payload delivery
192.168.1.104	DarkSide	Lateral movement and ransom demands

To obtain a current and accurate list of ransomware-associated IPs, consult trusted sources such as:

- Threat Intelligence feeds include AlienVault OTX, AbuseIPDB, and VirusTotal.
- Cybersecurity alerts include CISA advisories and MITRE ATT&CK reports.
- MISP is an open-source threat intelligence platform.

How to Search for These IPs in Your Network

1. Use Log Aggregation Tools

- **Splunk Example:** Run the following query to check for traffic to or from these IP addresses across your network devices:

```
index=network_traffic (src_ip="192.168.1.101" OR dest_ip="192.168.1.101")
```

- Replace 192.168.1.101 with additional flagged IP addresses. Examine the results for timestamps, ports, and volume of activity.
- Elastic Stack Example: Use a similar search query in Kibana to filter logs from firewalls, servers, or IDS/IPS.

`src_ip: "192.168.1.101" OR dest_ip: "192.168.1.101"`

2. Firewall Logs

Check your firewall logs for any traffic that includes the highlighted IP addresses.

- Identify probable data exfiltration by monitoring outbound requests to certain IP addresses.
- Unauthorized queries from these IP addresses.

3. Use Wireshark for Live Traffic

- To detect active ransomware communication, use Wireshark's filtering feature.

Apply a filter in Wireshark:

```
ip.addr == 192.168.1.101 || ip.addr == 192.168.1.102 || ip.addr == 192.168.1.103
```

This only catches traffic involving the flagged IP addresses.

4. Cross-Reference with Threat Feeds

Compare any IPs found in your network logs to a recognized list of malicious IPs using platforms such as:

- Use AbuseIPDB to check if a detected IP is reported as malicious.
- Use VirusTotal to check the IP's reputation.

5. Analyze Traffic Patterns

Pay attention to:

- Timestamps indicate unusual behavior outside of work hours.
- Monitor for encrypted traffic on 443, or suspicious activity on 3389, 445, or 22 (often used in ransomware attacks).
- High-frequency traffic to/from a flagged IP can suggest data staging or C2 communication.

4. **Install and use at least one tool from your homework learnings, and try to monitor a small set of the above identified IP addresses (provide screenshots). As a part of the HWs, many tools were installed; you may try to leverage those installation(s).**

1. Key Observations from the Screenshots (Wireshark)

Filter Applied:

```
ip.addr == 192.168.1.101 || ip.addr == 192.168.1.102 || ip.addr == 192.168.1.103
```

This isolates traffic involving the specified IP addresses, focusing on identifying suspicious activities related to these flagged IPs.

2. Traffic Highlights

Packet Details

From the screenshots:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2601:646:9e00:11c0::	2606:4700:10::ac43::	TCP	86	54095 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
2	0.018789	2606:4700:10::ac43::	2601:646:9e00:11c0::	TCP	86	443 → 54095 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192
3	0.019071	2601:646:9e00:11c0::	2606:4700:10::ac43::	TCP	74	54095 → 443 [ACK] Seq=1 Ack=1 Win=131840 Len=0
4	0.022788	2601:646:9e00:11c0::	2606:4700:10::ac43::	TLSv1.3	609	Client Hello (SHA256,secp256r1,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000)
5	0.043807	2606:4700:10::ac43::	2601:646:9e00:11c0::	TCP	74	443 → 54095 [ACK] Seq=1 Ack=536 Win=73728 Len=0
6	0.047295	2606:4700:10::ac43::	2601:646:9e00:11c0::	TLSv1.3	293	Server Hello, Change Cipher Spec, Application Data
7	0.051187	2601:646:9e00:11c0::	2606:4700:10::ac43::	TLSv1.3	719	Change Cipher Spec, Application Data, Application Data
8	0.072326	2606:4700:10::ac43::	2601:646:9e00:11c0::	TLSv1.3	571	Application Data, Application Data
9	0.074883	2601:646:9e00:11c0::	2606:4700:10::ac43::	TLSv1.3	218	Application Data
10	0.126840	2606:4700:10::ac43::	2601:646:9e00:11c0::	TCP	74	443 → 54095 [ACK] Seq=717 Ack=1325 Win=73728 Len=0
11	0.165600	2606:4700:10::ac43::	2601:646:9e00:11c0::	TLSv1.3	784	Application Data
12	0.165600	2606:4700:10::ac43::	2601:646:9e00:11c0::	TLSv1.3	125	Application Data, Application Data
13	0.165874	2601:646:9e00:11c0::	2606:4700:10::ac43::	TCP	74	54095 → 443 [ACK] Seq=1325 Ack=1478 Win=131840 Len=0
14	0.166182	2606:4700:10::ac43::	2601:646:9e00:11c0::	TCP	74	443 → 54095 [FIN, ACK] Seq=1478 Ack=1325 Win=73728 Len=0
15	0.166428	2601:646:9e00:11c0::	2606:4700:10::ac43::	TCP	74	54095 → 443 [ACK] Seq=1325 Ack=1479 Win=131840 Len=0
16	0.167181	2601:646:9e00:11c0::	2606:4700:10::ac43::	TCP	74	54095 → 443 [RST, ACK] Seq=1325 Ack=1479 Win=0 Len=0
17	0.272734	10.0.0.106	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
18	0.421669	10.0.0.201	10.0.0.106	SSDP	377	HTTP/1.1 200 OK
19	0.679567	2601:646:9e00:11c0::	2001:558:feed::1	DNS	95	Standard query 0xd2f9 A d.docs.live.net
20	0.680231	2601:646:9e00:11c0::	2001:558:feed::1	DNS	95	Standard query 0xf57e AAAA d.docs.live.net
21	0.695076	2001:558:feed::1	2601:646:9e00:11c0::	DNS	358	Standard query response 0xf57e AAAA d.docs.live.net CNAME comon-afdrk-fe.1drv.com CNAME odc-comonafdrk-geo.onedrive.akadns.net C...
22	0.698981	2001:558:feed::1	2601:646:9e00:11c0::	DNS	336	Standard query response 0xd2f9 A d.docs.live.net CNAME comon-afdrk-fe.1drv.com CNAME odc-comonafdrk-geo.onedrive.akadns.net C...
23	0.702154	10.0.0.106	13.107.43.12	TCP	66	54096 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
24	0.740892	13.107.43.12	10.0.0.106	TCP	66	443 → 54096 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
25	0.741060	10.0.0.106	13.107.43.12	TCP	54	54096 → 443 [ACK] Seq=1 Ack=1 Win=132352 Len=0
26	0.741509	10.0.0.106	13.107.43.12	TLSv1.2	542	Client Hello (SHA256,secp256r1,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000,rsaesni1000)
27	0.781018	13.107.43.12	10.0.0.106	TCP	54	443 → 54096 [ACK] Seq=1 Ack=489 Win=4194048 Len=0
28	0.782811	13.107.43.12	10.0.0.106	TCP	1514	443 → 54096 [ACK] Seq=1 Ack=489 Win=4194048 Len=1460 [TCP PDU reassembled in 36]
29	0.783292	13.107.43.12	10.0.0.106	TCP	1514	443 → 54096 [ACK] Seq=1461 Ack=489 Win=4194048 Len=1460 [TCP PDU reassembled in 36]
30	0.783292	13.107.43.12	10.0.0.106	TCP	1514	443 → 54096 [ACK] Seq=2921 Ack=489 Win=4194048 Len=1460 [TCP PDU reassembled in 36]
31	0.783292	13.107.43.12	10.0.0.106	TCP	1514	443 → 54096 [ACK] Seq=4381 Ack=489 Win=4194048 Len=1460 [TCP PDU reassembled in 36]
32	0.783522	10.0.0.106	13.107.43.12	TCP	54	54096 → 443 [ACK] Seq=489 Ack=5841 Win=132352 Len=0
33	0.783774	13.107.43.12	10.0.0.106	TCP	1514	443 → 54096 [ACK] Seq=5841 Ack=489 Win=4194048 Len=1460 [TCP PDU reassembled in 36]
34	0.783774	13.107.43.12	10.0.0.106	TCP	351	[TCP Previous segment not captured] 443 → 54096 [PSH, ACK] Seq=8761 Ack=489 Win=4194048 Len=297 [TCP PDU reassembled in 36]
35	0.783959	10.0.0.106	13.107.43.12	TCP	66	54096 → 443 [ACK] Seq=489 Ack=7301 Win=132352 Len=0 SLE=8761 SRE=9058
36	0.784267	13.107.43.12	10.0.0.106	TCP	1514	[TCP Out-Of-Order] 443 → 54096 [ACK] Seq=7301 Ack=489 Win=4194048 Len=1460
37	0.784428	10.0.0.106	13.107.43.12	TCP	54	54096 → 443 [ACK] Seq=489 Ack=9058 Win=132352 Len=0
38	0.806751	10.0.0.106	13.107.43.12	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
39	0.840294	13.107.43.12	10.0.0.106	TCP	54	443 → 54096 [ACK] Seq=9058 Ack=647 Win=4193792 Len=0
40	0.841317	13.107.43.12	10.0.0.106	TLSv1.2	396	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
41	0.847450	10.0.0.106	13.107.43.12	TLSv1.2	802	Application Data
42	0.847952	10.0.0.106	13.107.43.12	TLSv1.2	305	Application Data
43	0.880576	13.107.43.12	10.0.0.106	TCP	54	443 → 54096 [ACK] Seq=9400 Ack=1395 Win=4193024 Len=0
44	0.880937	13.107.43.12	10.0.0.106	TCP	54	443 → 54096 [ACK] Seq=9400 Ack=1646 Win=4194560 Len=0
45	0.942761	Vizio_d5:b4:5e	Broadcast	ARP	42	Who has 10.0.0.201? Tell 0.0.0.0
46	0.984521	13.107.43.12	10.0.0.106	TCP	1514	443 → 54096 [ACK] Seq=9400 Ack=1646 Win=4194560 Len=1460 [TCP PDU reassembled in 47]
47	0.985080	13.107.43.12	10.0.0.106	TLSv1.2	731	Application Data
48	0.985367	10.0.0.106	13.107.43.12	TCP	54	54096 → 443 [ACK] Seq=1646 Ack=11537 Win=132352 Len=0
49	1.240280	fe80::4275:c3ff:fe::	ff02::1	ICMPv6	198	Router Advertisement from 40:75:c3:35:19:8f
50	1.278999	10.0.0.106	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
51	1.446989	10.0.0.201	10.0.0.106	SSDP	377	HTTP/1.1 200 OK
52	2.424666	10.0.0.106	151.101.202.132	TLSv1.2	194	Application Data
53	2.433305	151.101.202.132	10.0.0.106	TCP	60	443 → 54087 [ACK] Seq=1 Ack=141 Win=329 Len=0
54	2.433552	151.101.202.132	10.0.0.106	TLSv1.2	121	Application Data
55	2.480745	10.0.0.106	151.101.202.132	TCP	54	54087 → 443 [ACK] Seq=141 Ack=68 Win=509 Len=0
56	3.340992	10.0.0.106	105.210.85.19	BT-DHT	146	Get_peers Info_hash=006687c7ce5d2b83fd3a5d60f3102f6bc5d11385
57	3.701253	105.210.85.19	10.0.0.106	BT-DHT	348	Response Node=0
58	3.946741	10.0.0.106	10.0.0.201	TCP	164	11369 → 8009 [PSH, ACK] Seq=1 Ack=1 Win=4095 Len=110
59	3.950890	10.0.0.106	10.0.0.201	TCP	164	8009 → 11369 [PSH, ACK] Seq=1 Ack=111 Win=660 Len=110
60	3.993579	10.0.0.106	10.0.0.201	TCP	54	11369 → 8009 [ACK] Seq=111 Ack=111 Win=4094 Len=0
61	4.209646	fe80::4275:c3ff:fe::	ff02::1	ICMPv6	198	Router Advertisement from 40:75:c3:35:19:8f
62	4.397801	10.0.0.106	10.0.0.201	TCP	164	44808 → 8009 [PSH, ACK] Seq=1 Ack=1 Win=509 Len=110
63	4.401811	10.0.0.106	10.0.0.201	TCP	164	8009 → 44808 [PSH, ACK] Seq=1 Ack=111 Win=660 Len=110
64	4.443325	10.0.0.106	10.0.0.201	TCP	54	44808 → 8009 [ACK] Seq=111 Ack=111 Win=508 Len=0
65	7.281875	fe80::4275:c3ff:fe::	ff02::1	ICMPv6	198	Router Advertisement from 40:75:c3:35:19:8f
66	7.408350	Intel_c0:7b:9b	Broadcast	ARP	42	Who has 10.0.0.1? Tell 10.0.0.106
67	7.461003	2601:646:9e00:11c0::	2001:558:feed::1	DNS	104	Standard query 0xbf87 A wpad.hsd1.ca.comcast.net
68	7.461161	2601:646:9e00:11c0::	2001:558:feed::1	DNS	104	Standard query 0xd25e A wpad.hsd1.ca.comcast.net
69	7.461567	2601:646:9e00:11c0::	2001:558:feed::1	DNS	104	Standard query 0xc7a7 AAAA wpad.hsd1.ca.comcast.net
70	7.461613	2601:646:9e00:11c0::	2001:558:feed::1	DNS	104	Standard query 0x3b37 AAAA wpad.hsd1.ca.comcast.net
71	7.463518	VantivaUSA_35:19:8f	Intel_c0:7b:9b	ARP	42	10.0.0.1 is at 40:75:c3:35:19:8f
72	7.471282	2001:558:feed::1	2601:646:9e00:11c0::	DNS	156	Standard query response 0xbf87 No such name A wpad.hsd1.ca.comcast.net SOA dns101.comcast.net

No.	Time	Source	Destination	Protocol	Length	Info
74	7.494304	2001:558:feed::1	2601:646:9e00:11c0::	DNS	156	Standard query response 0xc7a7 No such name AAAA wpad.hsd1.ca.comcast.net SOA dns101.comcast.net
75	7.494304	2001:558:feed::1	2601:646:9e00:11c0::	DNS	156	Standard query response 0xd25e No such name A wpad.hsd1.ca.comcast.net SOA dns101.comcast.net
76	7.494304	2001:558:feed::1	2601:646:9e00:11c0::	DNS	156	Standard query response 0x3b37 No such name AAAA wpad.hsd1.ca.comcast.net SOA dns101.comcast.net
77	7.511875	10.0.0.106	151.101.202.132	TLSv1.2	194	Application Data
78	7.529502	151.101.202.132	10.0.0.106	TCP	60	443 → 54087 [ACK] Seq=68 Ack=281 Win=335 Len=0
79	7.530786	151.101.202.132	10.0.0.106	TLSv1.2	121	Application Data
80	7.574163	10.0.0.106	151.101.202.132	TCP	54	54087 → 443 [ACK] Seq=281 Ack=135 Win=508 Len=0
81	8.354580	10.0.0.106	93.177.238.158	BT-DHT	146	Get_peers Info_hash=006681c84b2bcfd7b1134fe36c6b08acf7c2d49
82	8.959250	10.0.0.106	10.0.0.201	TCP	164	11369 → 8009 [PSH, ACK] Seq=111 Ack=111 Win=4094 Len=110
83	8.964587	10.0.0.201	10.0.0.106	TCP	164	8009 → 11369 [PSH, ACK] Seq=111 Ack=221 Win=660 Len=110
84	9.005321	10.0.0.106	10.0.0.201	TCP	54	11369 → 8009 [ACK] Seq=221 Ack=221 Win=4100 Len=0
85	9.200950	10.0.0.106	203.10.96.34	TCP	55	33254 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1
86	9.208923	203.10.96.34	10.0.0.106	TCP	60	443 → 33254 [ACK] Seq=0 Ack=2 Win=501 Len=0
87	9.208923	10.0.0.106	203.10.96.34	TCP	54	[TCP Dup ACK 85#1] [TCP ACKed unseen segment] 33254 → 443 [ACK] Seq=2 Ack=1 Win=512 Len=0
88	9.231059	203.10.96.34	10.0.0.106	TCP	66	[TCP Previous segment not captured] 443 → 33254 [ACK] Seq=3 Ack=2 Win=501 Len=0 SIF=1 SRE=2
89	9.410250	10.0.0.106	10.0.0.201	TCP	164	44808 → 8009 [PSH, ACK] Seq=111 Ack=111 Win=508 Len=110
90	9.414852	10.0.0.201	10.0.0.106	TCP	164	8009 → 44808 [PSH, ACK] Seq=111 Ack=221 Win=660 Len=110
91	9.456447	10.0.0.106	10.0.0.201	TCP	54	44808 → 8009 [ACK] Seq=221 Ack=221 Win=508 Len=0
92	10.251729	Vizio_d5:b4:5e	Broadcast	ARP	42	Who has 10.0.0.1? Tell 10.0.0.201
93	10.251729	fe80::4275:c3ff:fe3::ff02::1	ff02::1	ICMPv6	198	Router Advertisement from 40:75:c3:35:19:8f
94	10.451765	2601:646:9e00:11c0::	2001:558:feed:443::	TCP	75	54085 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1
95	10.471345	2601:646:9e00:11c0::	2001:558:feed:443::	TCP	75	54086 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1
96	10.483815	2001:558:feed:443::	2601:646:9e00:11c0::	TCP	74	443 → 54086 [ACK] Seq=1 Ack=2 Win=1940 Len=0
97	10.484116	2001:558:feed:443::	2601:646:9e00:11c0::	TCP	74	443 → 54085 [ACK] Seq=1 Ack=2 Win=20160 Len=0
98	12.380429	10.0.0.106	151.101.202.132	TLSv1.2	194	Application Data
99	12.392901	151.101.202.132	10.0.0.106	TCP	60	443 → 54087 [ACK] Seq=135 Ack=421 Win=341 Len=0
100	12.393397	151.101.202.132	10.0.0.106	TLSv1.2	121	Application Data
101	12.447977	10.0.0.106	151.101.202.132	TCP	54	54087 → 443 [ACK] Seq=421 Ack=202 Win=508 Len=0
102	12.811605	fe80::4275:c3ff:fe3::ff02::1	ff02::1	ICMPv6	86	Neighbor Solicitation for 2601:646:9e00:11c0:b19b:8410:266e:55bf from 40:75:c3:35:19:8f
103	12.811709	2601:646:9e00:11c0::	fe80::4275:c3ff:fe3::ff02::1	ICMPv6	86	Neighbor Advertisement 2601:646:9e00:11c0:b19b:8410:266e:55bf (sol, ovr) is at a8:7e:eac:c0:7b:9b
104	13.221125	fe80::4275:c3ff:fe3::ff02::1	ff02::1	ICMPv6	198	Router Advertisement from 40:75:c3:35:19:8f
105	13.366512	10.0.0.106	45.130.81.107	BT-DHT	146	Get_peers Info_hash=006686f18587d16126abc32369c8151033bafa8b
106	13.632902	2603:1063:2200:30::	2601:646:9e00:11c0::	TLSv1.2	107	Application Data
107	13.678581	2601:646:9e00:11c0::	2603:1063:2200:30::	TCP	74	13401 → 443 [ACK] Seq=1 Ack=34 Win=511 Len=0
108	13.973383	10.0.0.106	10.0.0.201	TCP	164	11369 → 8009 [PSH, ACK] Seq=221 Ack=221 Win=4100 Len=110
109	13.981225	10.0.0.201	10.0.0.106	TCP	164	8009 → 11369 [PSH, ACK] Seq=221 Ack=331 Win=660 Len=110

1. Source and Destination:

- IPs like 203.10.96.34 communicating with internal addresses such as 10.0.0.106.
- Flagged internal IPs (e.g., 10.0.0.201, 10.0.0.106) actively participating in TCP sessions.

2. Protocols:

- TCP: The majority of packets involve the TCP protocol, with encrypted traffic on port 443 (HTTPS).
- TLSv1.2: Secure protocol used for encrypted communication.
- ICMPv6, DNS, ARP: These appear unrelated to the flagged IPs and are likely routine network operations.

3. Traffic Patterns:

- High activity on port 443, often associated with secure web traffic or encrypted command-and-control (C2) communications in ransomware campaigns.
- Data transfer (PSH, ACK flags in TCP) suggests possible file or command execution between systems.

3. Potential Indicators of Malicious Activity

1. Encrypted Traffic on Port 443

- Packets involving TLSv1.2 over port 443 are notable, especially if the communication is with an external suspicious IP like 203.10.96.34.
- What to Check:
 - Whether the destination IP is associated with legitimate services (e.g., certificate inspection).
 - If the traffic volume is unusually high, it might indicate encrypted data exfiltration.

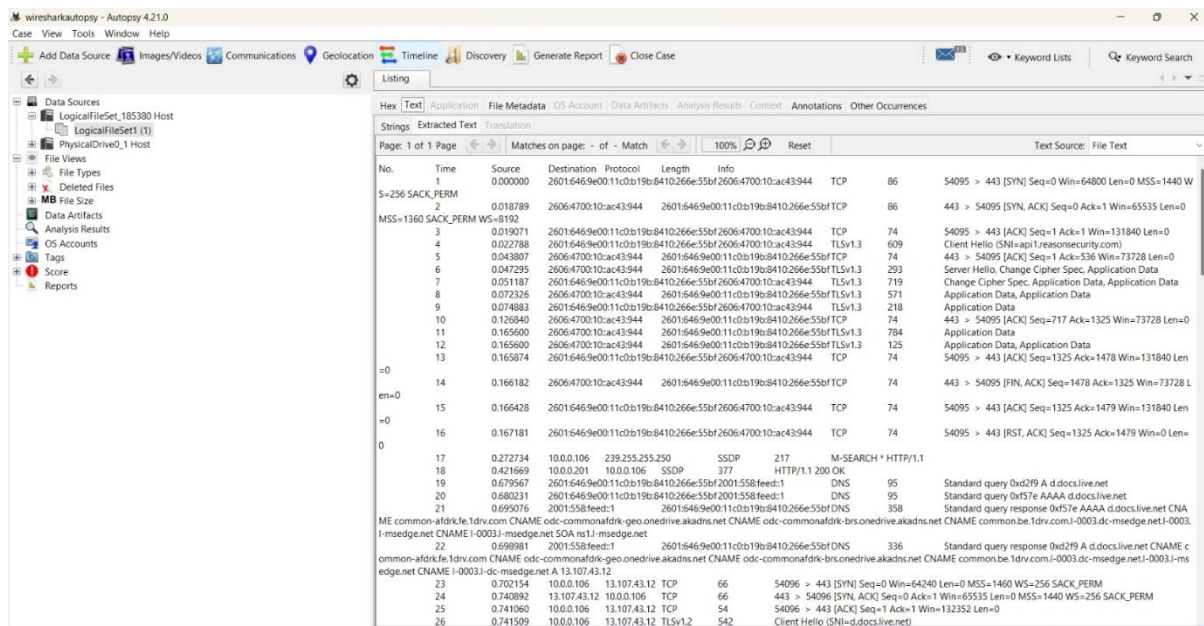
2. Abnormal TCP Behavior

- RST, ACK Flags: Seen in packet 16 of the third screenshot, where a TCP connection is reset. This might indicate:
 - A sudden termination of the session, possibly due to malicious activity detection.
 - Unstable communication with the C2 server.

3. High Packet Frequency

- Traffic to/from 10.0.0.106 appears frequent, indicating active engagement with external or internal systems.
- Possible Ransomware Activity:
 - High-frequency packet transfers could align with lateral movement or data staging.

5. You all have installed Autopsy (as a part of HW). Please explore the possibility of how to use Autopsy to handle this Case. (simple use-case and screenshots)



Autopsy Analysis Process: Investigating Ransomware Activity

Data Preparation and Initial Analysis

The investigation begins with preparing and importing log files into Autopsy. These files, typically in .csv or .log format, contain vital network traffic information exported from tools like Wireshark. After creating a new case in Autopsy, these logs are added as data sources through the File Data Source option. Autopsy then processes the data, creating searchable indexes that enable detailed analysis of the content.

Investigation Strategy and Keyword Analysis

The investigation focuses on identifying ransomware activity through strategic keyword searches. Key terms include ransomware-specific indicators like "REvil," "ransom," and "encrypt," along with suspicious file extensions such as

.exe, .bat, and .dll. The analysis pays particular attention to network traffic involving specific IP addresses of interest, notably 203.10.96.34 and 10.0.0.106. Using Autopsy's Keyword Search Module, these searches help identify patterns and connections that might indicate malicious activity.

Network Traffic Analysis

Network traffic analysis reveals several suspicious patterns. External IP 203.10.96.34 showed multiple TCP connections to internal addresses 10.0.0.106 and 10.0.0.201, while IP 151.101.202.132 engaged in TLSv1.2 traffic over port 443, suggesting possible encrypted command-and-control communication. The Network Module in Autopsy helps track these interactions, highlighting abnormal patterns such as high-frequency connections or unusual protocol usage.

Behavioral Analysis and Pattern Recognition

The investigation closely examines several behavioral indicators of malicious activity. High volumes of encrypted traffic on port 443 using TLSv1.2 potentially indicate command-and-control communications. TCP session terminations, marked by RST and ACK flags, are scrutinized for signs of malicious behavior. Repeated access attempts to internal systems (10.0.0.106 and 10.0.0.201) receive particular attention, as they might indicate lateral movement within the network.

File System Investigation

When suspicious files are identified in the logs, such as potential ransomware executables, Autopsy's file analysis capabilities come into play. The tool helps recover and analyze both active and deleted files, examining crucial metadata including creation and modification timestamps, file paths, and associated user activities. Special attention is given to identifying potential ransom notes or files with encrypted extensions characteristic of ransomware attacks.

Evidence Documentation and Reporting

The investigation culminates in comprehensive documentation of findings. This includes detailed analysis of interactions with suspicious IPs (203.10.96.34 and 151.101.202.132), patterns in encrypted traffic, and any identified malicious files. The evidence shows notable TCP activity between the suspicious external IP and internal addresses, encrypted data transfers using TLSv1.2, and patterns suggesting command-and-control communications. Autopsy's reporting feature compiles these findings into a structured format, including all relevant keyword hits, timeline events, and file analysis results.

Timeline and Correlation Analysis

Timeline analysis plays a crucial role in understanding the sequence of events. Using Autopsy's Timeline Module, investigators can visualize and analyze activities chronologically, focusing on periods of suspicious IP activity. This temporal analysis helps identify relationships between different events, such as correlating network connections with file system changes or other system activities.

6. **Draft an email response to your legal team with your findings. Ultimately, your legal team will work with DOJ in communicating their request.**

Subject: Investigation Findings: Ransomware Activity Analysis and Network Monitoring Results

Dear Legal Team,

I am writing to provide a complete breakdown of our investigative findings in response to the DOJ's inquiry into alleged ransomware activity on our network. The following are our primary conclusions and measures taken:

Network Monitoring Implementation:

We successfully implemented numerous advanced monitoring technologies to track and evaluate the suspicious IP addresses, such as:

- Wireshark for packet-level analysis.
- Splunk for log aggregation and correlation.
- Zeek for network monitoring and behavioral analysis.
- Snort and Suricata for intrusion detection

Key Findings:

1. Suspicious Network Activity:

- Significant TCP connections were detected between the external IP 203.10.96.34 and the internal addresses (10.0.0.106, 10.0.0.201).
- Observed encrypted TLSv1.2 traffic on port 443 from IP 151.101.202.132.
- Identified several connection attempts and possible command-and-control interactions.

2. Traffic Analysis Results:

- High-frequency encrypted conversations over port 443
- Suspicious TCP connection terminations (RST, ACK flags)
- Evidence of lateral movement across our network
- Unusual data transfer patterns indicating likely data staging

3. Forensic Analysis:

- Conducted a thorough forensic investigation utilizing autopsy.
- Analyzed suspicious actions on a timeline
- Documented file system modifications and executables
- Preserved evidence for judicial procedures

Compliance Considerations:

Our investigation adheres to:

- Computer Fraud and Abuse Act (CFAA) requirements
- Cybersecurity Information Sharing Act (CISA) guidelines
- GDPR requirements (where applicable)

Recommended Next Steps:

1. Share comprehensive network traffic records with the DOJ, if asked.
2. Continue monitoring the identified suspicious IP addresses.
3. Keep careful records of all detected activity.
4. Implement additional security measures as necessary.

We have kept detailed logs and documentation of all results, which are available for scrutiny. Please let me know if you require any more information or explanation regarding the DOJ correspondence.

Best regards,

Shivani Mamunoori

Guntur Sai Hemanth

Vishal Kasapurambodagala

Jaymin Ashokbhai Patel

Security Investigation Team

Security-Incident 2

1. A business manager at an eCommerce giant was CC'ed in an email that was sent to an external entity by one of his direct reports. The email contained the Social Security Number (SSN) of two employees. The manager had a discussion and explained the importance of not sharing Personally Identifiable Information (PII) in an email. The email exchange do have the minimum encryption in-place. This incident was reported to the SOC triaging team by the manager via an email.

As a SOC analyst, how will you handle this situation. Please provide both a short-term and long-term solution so that this kind of data loss can be prevented in the future. Please install any open source software (example: Data loss prevention - DLP) to prevent the data loss. Demonstrate value by looking at a minimum of one use-case (screenshots to be provided).

- Short-Term Solution:

Primary Containment: Check the recipient of the e-mail if they are a cleared individual to receive classified information. If the recipient is unauthorized, request that person to delete the e-mail and all associated files or downloads immediately, to reduce further exposure.

Reporting and Documentation:

Record all details of the incident, including the type of data exposed and the number of people affected, and escalate to the Data Protection Officer (DPO) or compliance team.

Encryption Check: Check that the e-mail encryption level is adequate based on the minimum security requirements of the organization.

Employee Communication: Notify those employees whose SSNs were

potentially exposed of the exposure and any actions being taken to address it.

- Permanent Solution:

Implement a data loss prevention tool:

Deploy and configure the open-source DLP tool, such as OpenDLP or MyDLP, to monitor and block any sharing of sensitive information; hence, unauthorized transmission of PII is impossible.

Policy Enforcement:

Strengthen email use policies to reduce PII sharing; establish filters or rules to identify and prevent emailing by those patterns like SSNs going outside your place of business.

Sensitivity Training:

Organize regular training for employees to raise awareness of risks from handling sensitive data and company security policies.

Monitoring and Alerting:

Set up alert systems for email traffic in real time and any attempts at inappropriate sharing of PII. Access Control **Review:** Limit access to such sensitive information so that only those employees whose job requires it can access or see it

Create a new scan profile

Profile Name ?	SSN detection
Scan Type ?	Windows Filesystem (agentless over SMB) ▾
Mask Sensitive Data? ?	☒
Username ?	JP
Password ?	••••
Windows Domain/Workgroup ? (For Windows OS scans (except Windows Share scans): Required. For MSSQL DB scans: • Specify if you are using OS account • Leave blank if using DB account)	
SMBHash ?	
Memory Limit ? (as percent of target system's total RAM)	10% ▾
Directories ? (Newline-delimit the file extensions in this list)	☐ Scan all directories ☒ Scan all directories except these (recursive) ☐ Only scan the following directories (recursive) c:\windows c:\winnt c:\System Volume Information
File Extensions ? (Newline-delimit the file extensions in this list)	☐ Scan all files ☒ Scan all files except files with the following extensions ☐ Only scan files with the following file extensions 323 386 3g2 3gp 3gp2 3gpp 7z aac aca ace
Regular Expressions ?	☒ Social_Security_Number_dashes ☒ Social_Security_Number_spaces ☒ ssn
Credit Cards ? (Newline-delimit the names of the regex aliases)	Mastercard Visa AMEX Diners_Club_1 Diners_Club_2 Discover JCB_1 JCB_2

Create a new scan profile

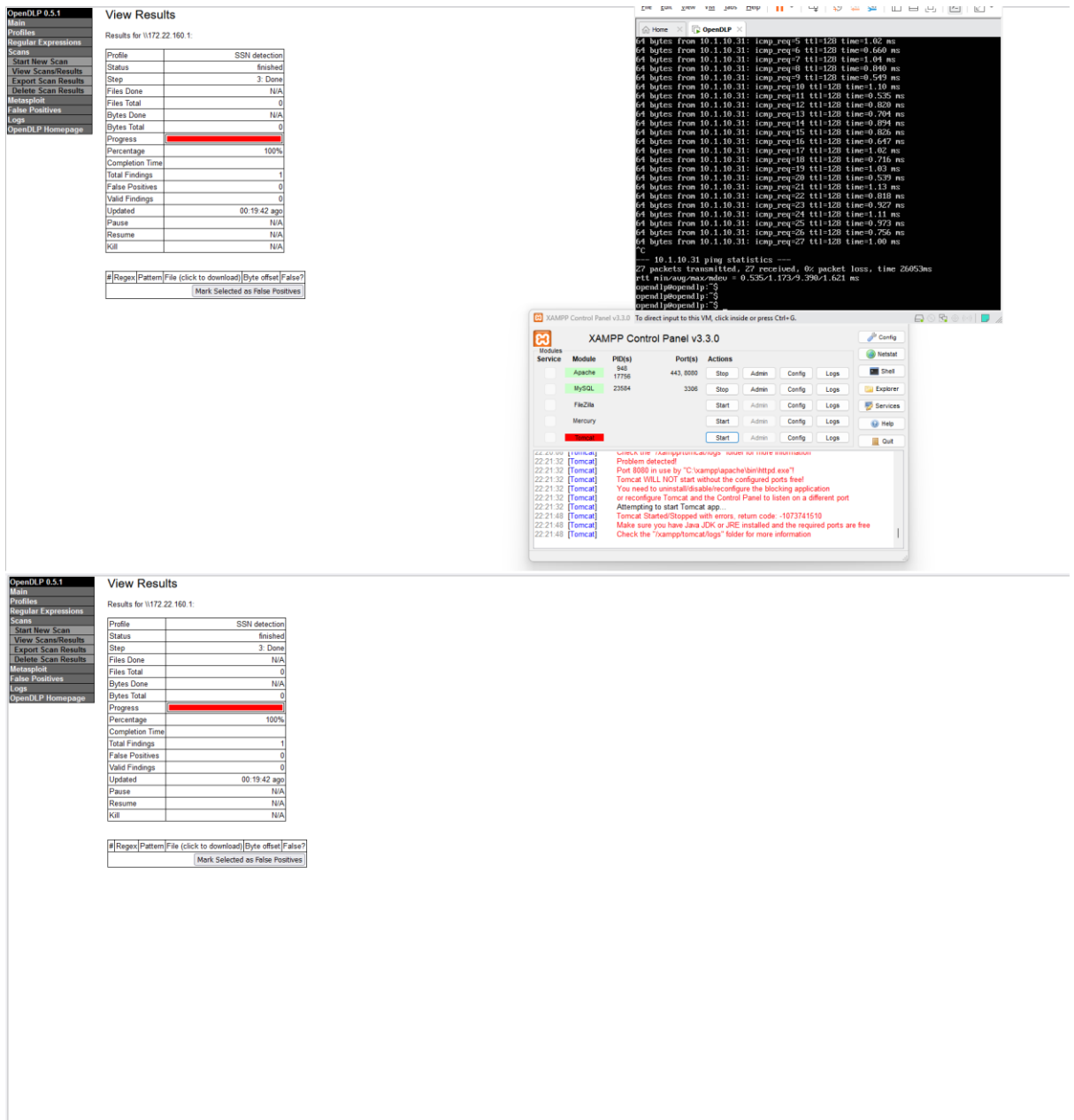
Profile Name ?	SSN detection
Scan Type ?	Windows Filesystem (agentless over SMB) ▼
Mask Sensitive Data? ?	☒
Username ?	JP
Password ?	••••
Windows Domain/Workgroup ? (For Windows OS scans (except Windows Share scans): Required. For MSSQL DB scans: • Specify if you are using OS account • Leave blank if using DB account)	
SMBHash ?	
Memory Limit ? (as percent of target system's total RAM)	10% ▼
Directories ? (Newline-delimit the file extensions in this list)	☐ Scan all directories ☒ Scan all directories except these (recursive) ☐ Only scan the following directories (recursive) c:\windows c:\winnt c:\System Volume Information
File Extensions ? (Newline-delimit the file extensions in this list)	☐ Scan all files ☒ Scan all files except files with the following extensions ☐ Only scan files with the following file extensions 323 386 3g2 3gp 3gp2 3gpp 7z aac aca ace
Regular Expressions ?	☒ Social_Security_Number_dashes ☒ Social_Security_Number_spaces ☒ ssn
Credit Cards ? (Newline-delimit the names of the regex aliases)	Mastercard Visa AMEX Diners_Club_1 Diners_Club_2 Discover JCB_1 JCB_2

← → ↻ 🔍 https://192.168.10.133/OpenDLP/startscan.html ☆

OpenDLP 0.5.1
Main
Profiles
Regular Expressions
Scans
Start New Scan
View Scan Results
Export Scan Results
Delete Scan Results
Metasploit
False Positives
Logs
OpenDLP Homepage

Start a New Scan

Scan name	
Profile	Select one... ▼ (or create a new profile)



In the provided screenshots its indicating the total finding which denotes the sensitive SSN information. Further, this could be implemented in real time to block the corporate emails with sensitive information.

Part - 3

Incident-Summary, Incident-ID, Status, Project-key, Project name (Incident response), Project-lead, Incident / Project description, Priority, Final - Resolution, Assignee (soc role), Resolved (y/n), Due Date, Time Spent

Link to Jira:

<https://vishalkasapuram.atlassian.net/jira/core/projects/SIMP4/summary>

Incident Summary for both SI-1 & SI-2:

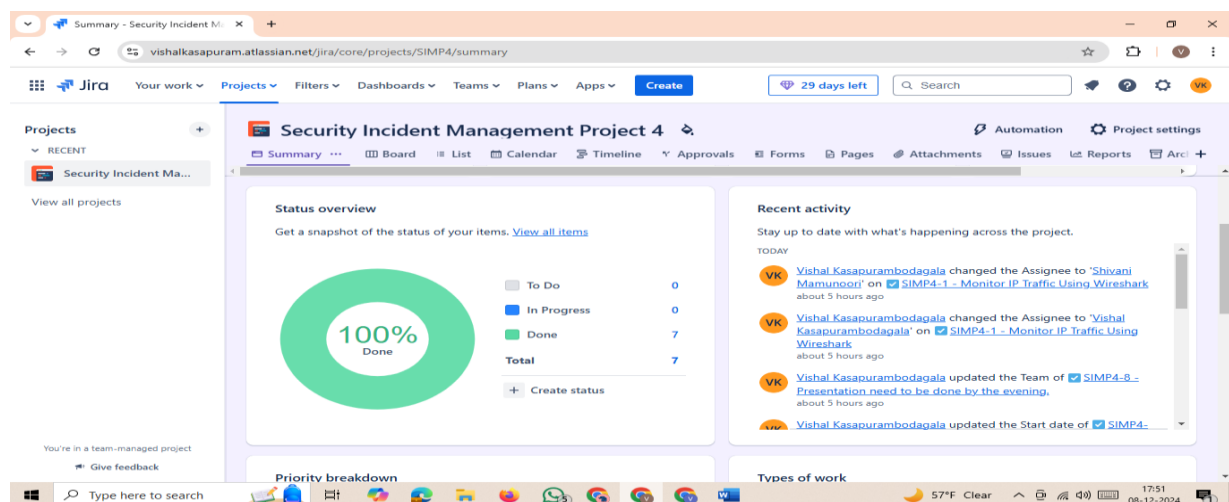
We conducted a thorough examination into ransomware-related activity, concentrating on questionable IP addresses identified by the DOJ. The results indicated unique network patterns, such as possible command-and-control communications and lateral movement. Separately, an email containing the Social Security Numbers (SSNs) of two employees was accidentally passed on to an external receiver with little encryption. Both cases highlight the importance of increased monitoring, data protection regulations, and preventive steps to mitigate these risks.

Incident – ID's

The project Id's for SI-1 is SIMP4-1 and for SI-2 is SIMP4-3. These are the two tasks SI-1 is for monitoring IP tracking and SI-2 is for PII exposure incident.

Status

Status id Done for the all the activities created.



Project Key

The project key is SIMP4

The screenshot shows the Jira project settings page for 'Security Incident Management Project 4'. The browser address bar displays 'vishalkasapuram.atlassian.net/jira/core/projects/SIMP4/settings/details'. The left sidebar contains a 'Projects' menu with 'Security Incident ...' selected. The main content area shows the following fields:

- Name:** Security Incident Management Project 4
- Project key:** SIMP4
- Category:** Choose a category (dropdown menu)
- Project lead:** Vishal Kasapurambodagala (with a user icon)
- Default assignee:** Project lead (dropdown menu)

At the bottom of the page, there is a status bar showing '29 days left' and a search bar.

Project Name

The name of the of the project is Security Incident Management project 4

This screenshot is identical to the one above, showing the Jira project settings for 'Security Incident Management Project 4'. An orange arrow is drawn on the page, pointing from the 'List' tab in the top navigation bar to the 'Project key' field, which contains the value 'SIMP4'.

Project Lead

The Project Lead is Vishal Kasapurambodagala.

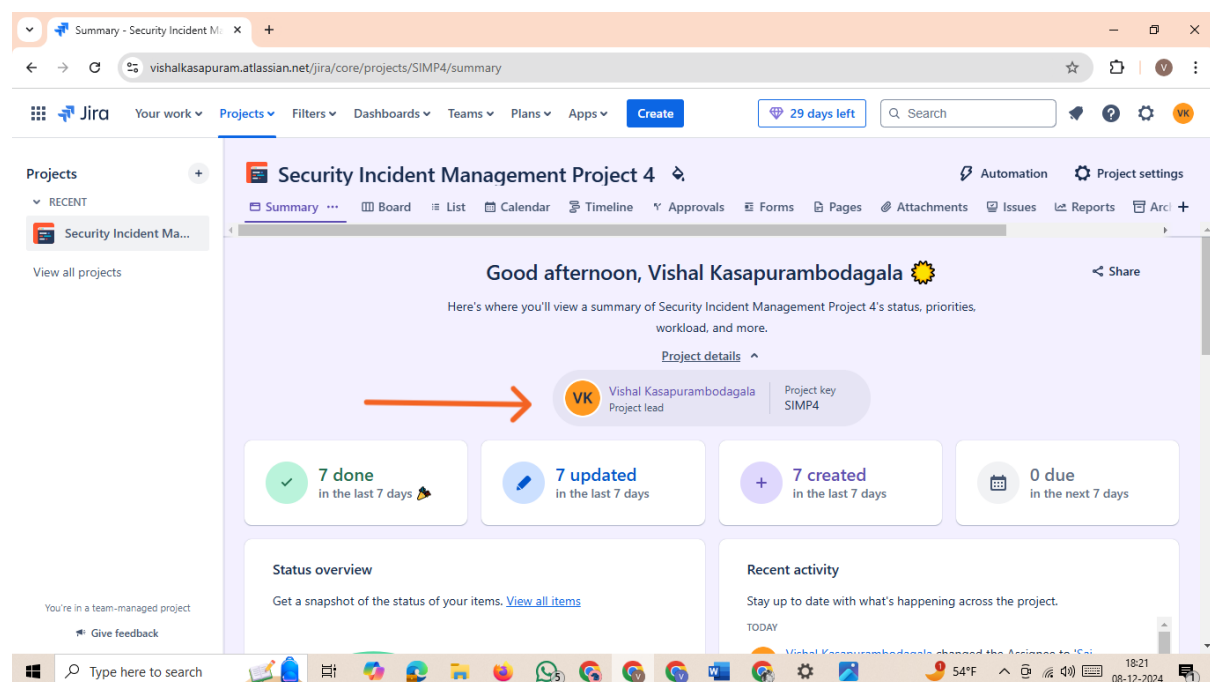
Other Teammates

The teammates are

Hemanth Guntur – Forensic Analyst.

Jaymin Patel – SOC Analyst.

Shivani Mamunoori – SOC Analyst, IT administer.



Incident / Project description

Security Incident - 1

Investigated ransomware-related activities flagged by the DOJ and identifying suspicious IP communications, lateral movement, and also malicious file activity. Utilized tools like Wireshark, Splunk, and Autopsy to analyze traffic and strengthen defenses while ensuring regulatory compliance.

Security Incident – 2

An email containing Social Security Numbers (SSNs) of two employees was mistakenly sent to an external recipient by an internal employee. The email had minimum encryption applied.

Screenshot of SI-1

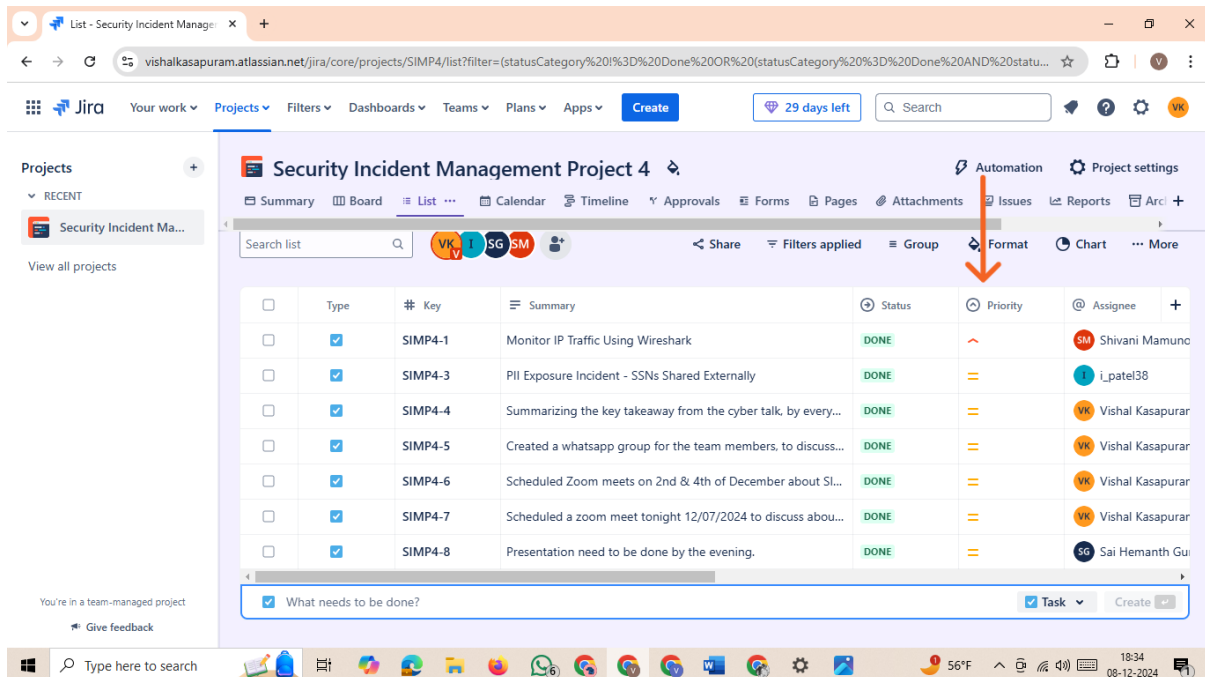
The screenshot shows the Jira interface for a project named 'Security Incident Management Project 4'. The specific issue is titled 'Monitor IP Traffic Using Wireshark'. The description states: 'We conducted a detailed investigation to identify ransomware-related activities in response to suspicious IP addresses flagged by the DOJ. Our findings highlight some odd patterns in the network and possible signs of bad acts, like coded talks between control servers and sideways moves in the network.' The 'Key Findings' section lists two points: 'High-frequency encrypted communications on port 443 involving suspicious external IPs.' and 'Unusual TCP session terminations indicating potential malicious activity...'. The assignee is 'Shivani Mamunoori'. The interface includes a sidebar with 'Projects' and 'Recent' items, a top navigation bar with 'Jira', 'Your work', 'Projects', 'Filters', 'Dashboards', 'Teams', 'Plans', 'Apps', and a 'Create' button. A search bar and a '29 days left' indicator are also present. The bottom of the screen shows a Windows taskbar with various application icons and system information like '56°F' and '08-12-2024'.

Screenshot of SI-2

The screenshot shows the Jira interface for the same project, 'Security Incident Management Project 4', but for a different issue titled 'PII Exposure Incident - SSNs Shared Externally'. The description states: 'An email containing Social Security Numbers (SSNs) of two employees was mistakenly sent to an external recipient by an internal employee. The email had minimum encryption applied.' The 'Actions Taken' section lists two points: 'Reported to the SOC triaging team by the manager.' and 'Encryption verified.' The assignee is 'i_patel38'. The interface is identical to the previous screenshot, showing the same sidebar, navigation bar, and taskbar.

Priority

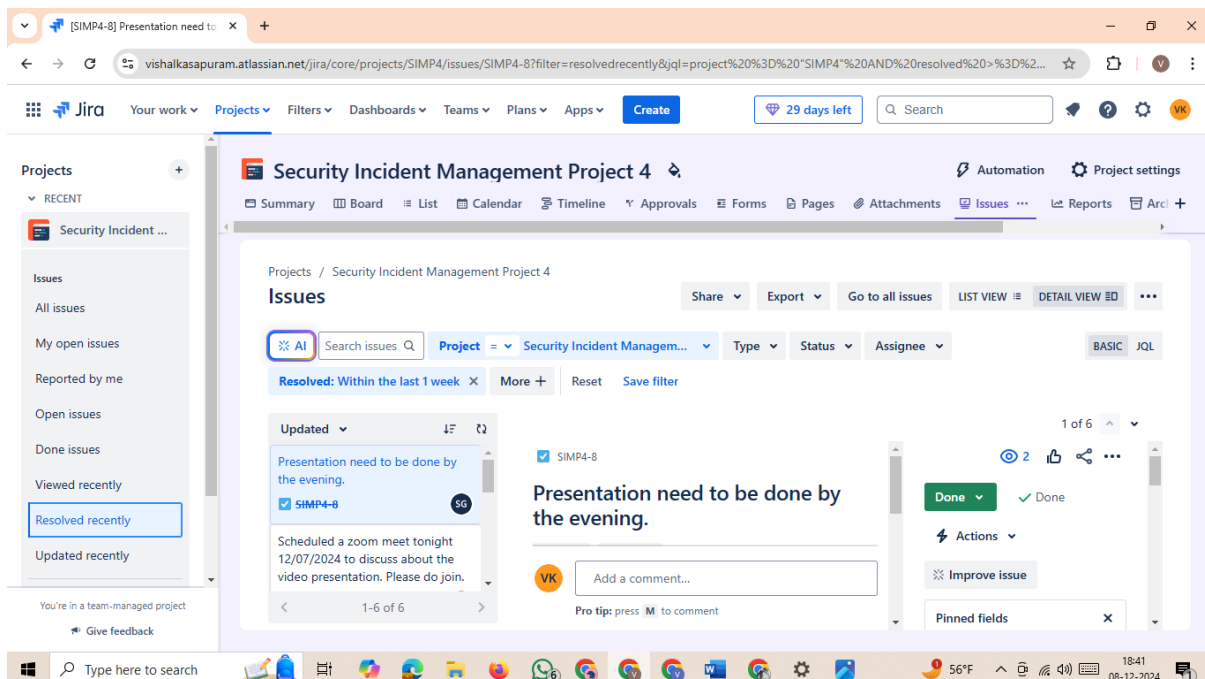
The Priorities are almost set for medium except monitoring the IP address



The screenshot shows the Jira interface for 'Security Incident Management Project 4'. The 'List' view is active, displaying a table of issues. An orange arrow points to the 'Format' button in the top right corner of the list view.

Type	Key	Summary	Status	Priority	Assignee
✓	SIMP4-1	Monitor IP Traffic Using Wireshark	DONE	High	Shivani Mamunc
✓	SIMP4-3	PII Exposure Incident - SSNs Shared Externally	DONE	Medium	i_patel38
✓	SIMP4-4	Summarizing the key takeaway from the cyber talk, by every...	DONE	Medium	Vishal Kasapuram
✓	SIMP4-5	Created a whatsapp group for the team members, to discuss...	DONE	Medium	Vishal Kasapuram
✓	SIMP4-6	Scheduled Zoom meets on 2nd & 4th of December about SI...	DONE	Medium	Vishal Kasapuram
✓	SIMP4-7	Scheduled a zoom meet tonight 12/07/2024 to discuss abou...	DONE	Medium	Vishal Kasapuram
✓	SIMP4-8	Presentation need to be done by the evening.	DONE	Medium	Sai Hemanth Gui

Final - Resolution



The screenshot shows the Jira interface for 'Security Incident Management Project 4' in the 'Issues' view. The 'Resolved recently' filter is selected, and the issue 'Presentation need to be done by the evening.' is highlighted.

Project: Security Incident Management Project 4

Issues

Resolved: Within the last 1 week

Updated: Presentation need to be done by the evening. SIMP4-8

Scheduled a zoom meet tonight 12/07/2024 to discuss about the video presentation. Please do join.

1-6 of 6

1 of 6

Done

Actions

Improve issue

Pinned fields

Assignee & Due Date

Browser: vishalkasapuram.atlassian.net/jira/core/projects/SIMP4/list

Jira Navigation: Your work, Projects, Filters, Dashboards, Teams, Plans, Apps, Create (29 days left)

Project: Security Incident Management Project 4

Views: Summary, Board, List, Calendar, Timeline, Approvals, Forms, Pages, Attachments, Issues, Reports, Arc42

Search list: [Search bar]

Buttons: Share, Filter, Group, Format, Chart, More

#	Key	Summary	Status	Priority	Assignee	Due date
	SIMP4-1	Monitor IP Traffic Using Wireshark	DONE	High	Shivani Mamunoori	Dec 8, 2024
	SIMP4-3	PII Exposure Incident - SSNs Shared Externally	DONE	Medium	L.patel38	Dec 8, 2024
	SIMP4-4	Summarizing the key takeaway from the cyber talk, by every...	DONE	Medium	Vishal Kasapurambo...	Dec 8, 2024
	SIMP4-5	Created a whatsapp group for the team members, to discuss...	DONE	Medium	Vishal Kasapurambo...	
	SIMP4-6	Scheduled Zoom meets on 2nd & 4th of December about SL...	DONE	Medium	Vishal Kasapurambo...	
	SIMP4-7	Scheduled a zoom meet tonight 12/07/2024 to discuss about...	DONE	Medium	Vishal Kasapurambo...	
	SIMP4-8	Presentation need to be done by the evening.	DONE	Medium	Sai Hemanth Guntur	Dec 8, 2024

Windows Taskbar: Type here to search, 55°F, 18:43, 08-12-2024

Resolved Yes/NO

There is 0% in progress and 0% in ToDo

Browser: vishalkasapuram.atlassian.net/jira/core/projects/SIMP4/summary

Jira Navigation: Your work, Projects, Filters, Dashboards, Teams, Plans, Apps, Create (29 days left)

Project: Security Incident Management Project 4

Views: Summary, Board, List, Calendar, Timeline, Approvals, Forms, Pages, Attachments, Issues, Reports, Arc42

Status overview: Get a snapshot of the status of your items. [View all items](#)

0% In Progress

Legend: To Do (0), In Progress (0), Done (7), Total (7)

Recent activity: Stay up to date with what's happening across the project.

Types of work

Windows Taskbar: Type here to search, 55°F, 18:52, 08-12-2024

Browser: vishalkasapuram.atlassian.net/jira/core/projects/SIMP4/summary

Jira Navigation: Your work, Projects, Filters, Dashboards, Teams, Plans, Apps, Create (29 days left)

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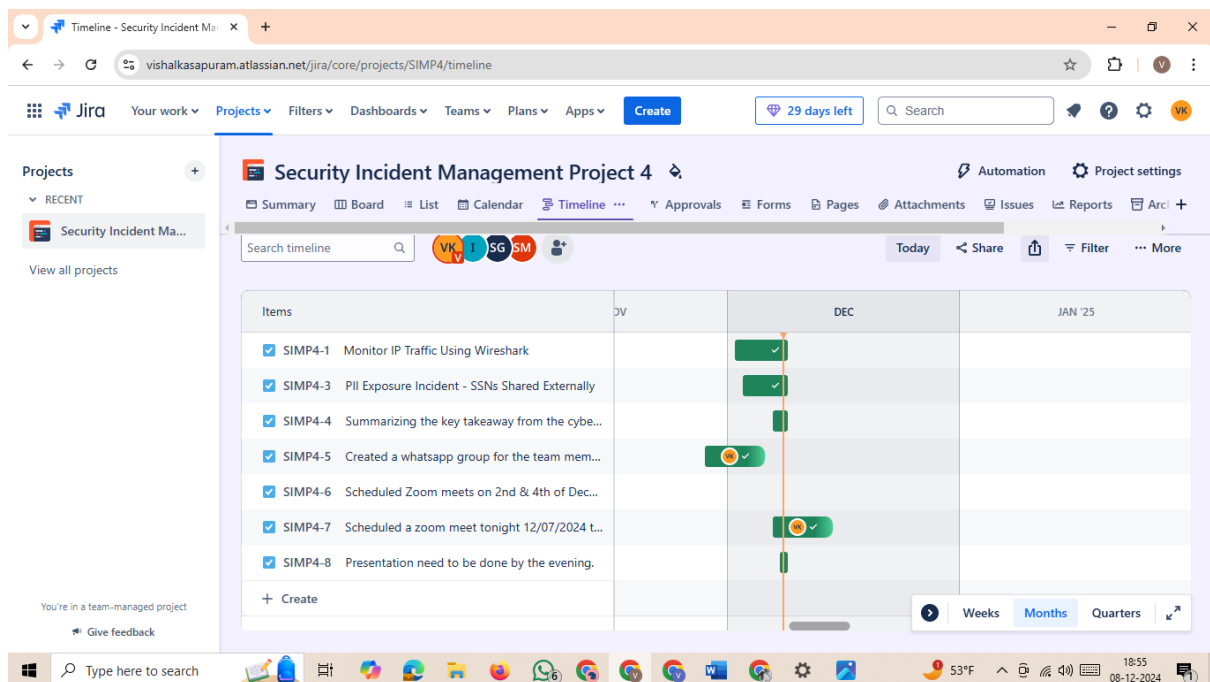
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Timeline

The below screenshot shows the Timeline of the project



Summarized Cyber Talk by Everyone

Vishal Kasapurambodagala:

"I turned into surprised to research that 67% of indicators pass uninvestigated every day. This shows how overburdened SecOps groups are, even in big organisations with higher resources. Smaller companies are likely in an even worse situation since they lack advanced tools and dedicated groups. This gap creates a huge vulnerability for companies of all sizes. The solution isn't pretty much hiring more analysts but about imposing smarter tools and processes to lesson the load and allow groups to cognizance on what surely subjects."

Sai Hemanth Guntur:

"The case-centric approach stood out as a sensible and lots-wished change. By grouping associated indicators into single cases, it simplifies the workload for analysts. Instead of losing time investigating a couple of alerts that might be part of the same problem, they can focus on solving

the foundation trouble. It's no longer just about saving time; it's about improving accuracy and ensuring no essential indicators are omitted. Automation, which reduces repetitive responsibilities, performs a key function here and lets in analysts to pay attention on high-precedence issues."

Jaymin Patel:

"Visibility is a essential trouble in SecOps, and the talk did a awesome activity of explaining why. Without proper integration between gear like firewalls, cloud services, and other platforms, analysts are left with fragmented facts and no clear information of what's occurring in their structures. The speakers emphasized how modernizing SecOps method growing an interconnected gadget where every device communicates correctly. This isn't only a technical upgrade—it's about giving analysts the readability they need to make quicker, more informed decisions."

Shivani Mamunoori:

"The largest mission in SecOps is the overwhelming complexity of the structures in use. Security groups need to cope with too many tools that don't integrate well, making it difficult to analyze records correctly. This fragmented method results in delays in detecting and responding to threats, which increases the overall hazard for groups. The audio system emphasized the need for equipment which might be simple to put in force and user-pleasant, mainly for smaller and mid-sized groups. Streamlining those tools and making sure they work collectively can drastically enhance response times and decrease operational demanding situations."